

DWG NO: WOK_1_2_Bonding_Head_TCB

NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:

- 1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
- 1.2 3D CAD DATASET (WOK_1_2_Bonding_Head_TCB.STEP) GOVERNS THE NOMINAL GEOMETRY FOR ALL UNTOLERANCED SURFACES.
- 1.3 GENERAL TOLERANCES: LINEAR +/- 0.05 mm, ANGULAR +/- 0.2 DEG (PER ISO 2768-f).

2. MATERIAL & ADDITIVE MANUFACTURING SPECIFICATION:

- 2.1 PRIMARY MATERIAL (FOR HIGH-TEMP TCB & SUB-MICRON THERMAL COPLANARITY):
TITANIUM ALLOY Ti-6Al-4V ELI (GRADE 23) CONFORMING TO ASTM F3001 / AMS 4930.
- CALCULATED FINISHED MASS: 85.1 g (DENSITY = 4.43 g/cm³).
- 2.2 SECONDARY MATERIAL (FOR LOW-THERMAL / ULTRA-HIGH DYNAMICS APPLICATIONS ONLY):
ALUMINUM ALLOY AISi10Mg CONFORMING TO ASTM F3318 / AMS 4880.
- CALCULATED FINISHED MASS: 51.5 g (DENSITY = 2.68 g/cm³).
- 2.3 FABRICATION PROCESS: LASER POWDER BED FUSION (PBF-LB / SLM).
- 2.4 LAYER THICKNESS: 30 TO 50 MICRONS.
- 2.5 INTERNAL MONOCOQUE CORE: 4-WAY PHASE-ALIGNED GYROID TPMS INFILL (T = 10.0 mm).

3. THERMAL PROCESSING & POWDER EVACUATION:

- 3.1 STRESS RELIEF VACUUM ANNEAL PRIOR TO BASEPLATE CUTTING:
 - FOR PRIMARY Ti-6Al-4V ELI: 680°C +/- 10°C FOR 2 HOURS UNDER HIGH VACUUM
 (< 1.0E-4 mbar) OR HIGH-PURITY ARGON, CONTROLLED FURNACE COOL.
 - FOR SECONDARY AISI10Mg: 300°C +/- 10°C FOR 2 HOURS UNDER INERT ATMOSPHERE.
- 3.2 100% UNMELTED POWDER EVACUATION: EXTRACTED VIA DUAL INDEPENDENT DIA 3.50 mm
 DEPOWDER PORTS USING ULTRASONIC CAVITATION AND PULSED DRY NITROGEN.
- 3.3 RESIDUAL POWDER VERIFICATION: GRAVIMETRIC DRIFT < 0.02 g OR INDUSTRIAL X-RAY CT.
- 3.4 SEAL PORTS FLUSH USING DIA 5.50 mm MATCHING ALLOY PLUGS SECURED WITH VACUUM-RATED
 ADHESIVE (LOCTITE 243) OR ORBITAL MICRO-TIG HERMETIC SEALING.

4. PNEUMATIC INTEGRITY (HERMETIC CORE):

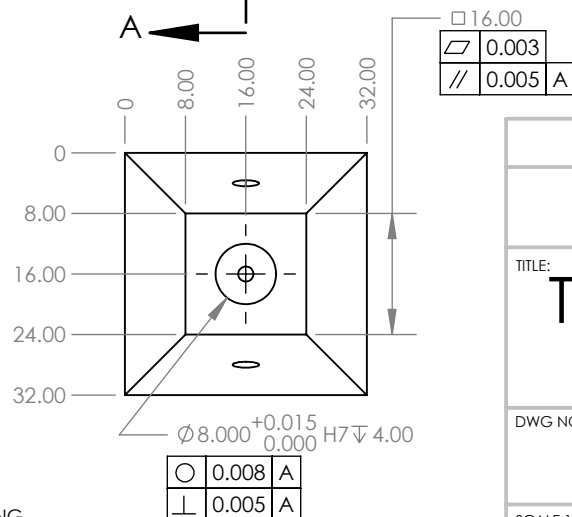
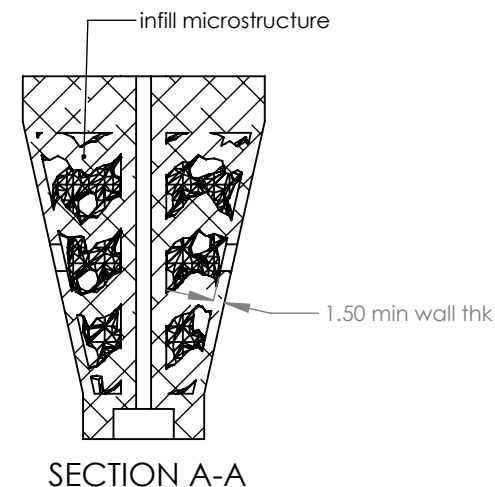
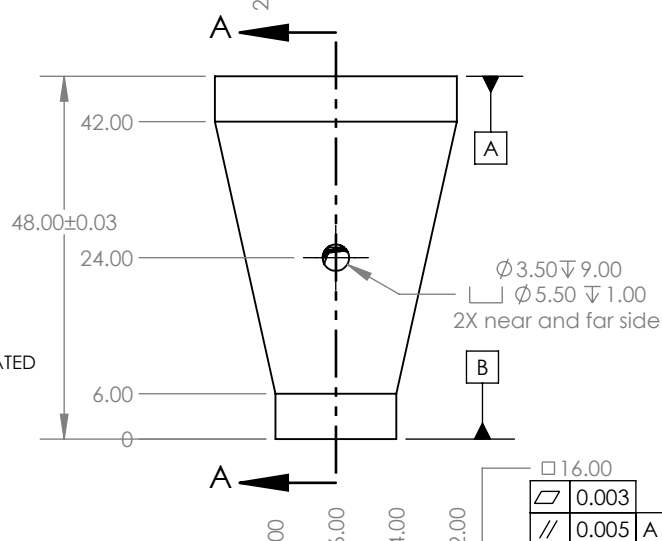
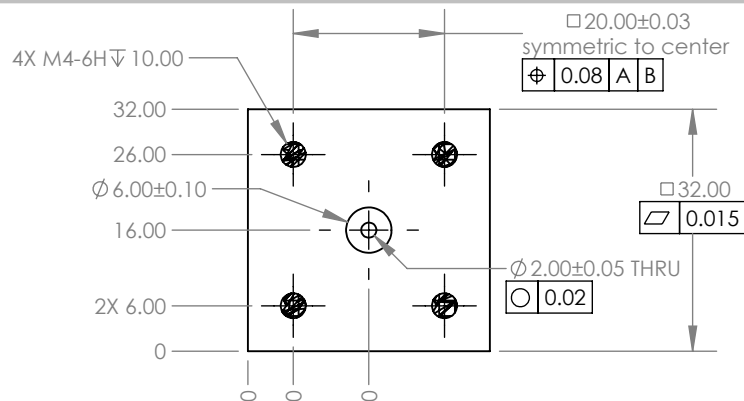
- 4.1 CENTRAL VACUUM CONDUIT (DIA 2.00 mm THRU) AND ISOLATED SLEEVE (DIA 6.00 mm) SHALL MAINTAIN 100% BULK HERMETIC SEALING WITHOUT VOIDS, CRACKS, OR PENETRATION.
- 4.2 VACUUM LEAKAGE RATE: SHALL NOT EXCEED 1.0×10^{-6} mbar*L/s UNDER -95 kPa DIFFERENTIAL PRESSURE (VERIFIED VIA ASTM E499 HELIUM MASS SPECTROMETRY LEAK TEST).

5. PRECISION FINISHING (SCHEME C TOOLING INTERFACE):

- 5.1 TOP MOUNTING FLANGE (DATUM A): PRECISION FINISH-MILLED, FLATNESS ≤ 0.015 mm, Ra ≤ 0.8 μ m.
5.2 BOTTOM COLLET BORE (DIA 8.000 H7) & END FACE (DATUM B):
HIGH-SPEED REAMED AND LAPPED (SCHEME C CLAMPING - ZERO LATERAL THREADS);
CYLINDRICITY ≤ 0.005 mm, END FACE FLATNESS ≤ 0.003 mm, Ra ≤ 0.4 μ m.
5.3 4X M4-6H TOP MOUNTING MATRIX: TAPPED 10.0 mm MIN FULL THREAD (PRE-DRILLED DIA 3.30 mm).
5.4 BREAK ALL SHARP EDGES 0.1 - 0.2 mm; MAINTAIN UNIFORM 1.50 mm MIN MONOCOQUE WALL.

6. SURFACE TREATMENT & CLEANROOM COMPLIANCE (ISO 14644-1 CLASS 3):

- 6.1 COATING / PASSIVATION:
- FOR Ti-6Al-4V ELI: TITANIUM ANODIZE TYPE II (ANTI-GALLING) PER AMS 2488 OR NITRIC ACID CHEMICAL PASSIVATION PER ASTM F86 (ZERO PARTICULATE FLAKING).
 - FOR AISi10Mg: HIGH-PHOSPHORUS ELECTROLESS NICKEL PLATING (ENP) PER MIL-C-26074 (5 - 8 um).
- 6.2 FINAL DECONTAMINATION: PRECISION ULTRASONIC CLEANING IN MULTI-STAGE DEIONIZED WATER (18 MOhm*cm) FOLLOWED BY CLASS 100 CLEAN-AIR DRYING AND DOUBLE VACUUM PACKAGING.



DO NOT SCALE DRAWING	REVISION:
TITLE: TCB DIE BONDING HEAD	
DWG NO. W1-SEMI-BE-TCB_RevD	A4
SCALE:1:1	SHEET 1 OF 1